

XC Training Guidelines

Our Philosophy

Distance running is not about one great workout or one great race. It is about accumulating hundreds of good training days over multiple seasons. The athletes who improve the most are usually not the ones who train the hardest for a week; they are the ones who train consistently, stay healthy, recover well, and trust the process.

Our program is built around the idea that every workout has a purpose. Some runs build your aerobic engine, some improve running economy and mechanics, some develop speed and power, and some simply help you recover so that the next hard workout is effective. No single workout makes you fast. Fitness is the result of weeks and months of training layered together.

We also train by time rather than mileage whenever possible. A 45-minute run is a similar aerobic stimulus whether one athlete covers 5 miles and another covers 7 miles. This allows runners of different abilities to train together while still receiving the appropriate workload.

The emphasis is on developing good habits, learning proper training principles, and progressing gradually rather than chasing mileage.

The Four Pillars of Training

1. Consistency

The single biggest predictor of improvement is consistency. Missing one workout rarely matters; missing weeks of training does. The goal is to stay healthy, show up regularly, and keep building fitness over time.

2. Easy Running

Easy running builds the aerobic engine that supports everything else we do. It should feel comfortable, conversational, and relaxed. If you cannot hold a conversation, you are probably running too fast.

A general guideline is goal 5K pace + 1:30–2:30 per mile, and heart rate should generally stay below 180 minus your age. Easy runs should leave you feeling like you could have continued running.

3. Quality Work

We typically perform no more than two quality sessions per week. Quality sessions include tempo runs, VO₂ max intervals, hill workouts, and races.

Hard workouts are effective because easy days stay easy.

4. Recovery

Recovery is part of training, not a break from training. Sleep, hydration, nutrition, mobility work, easy pacing, and strength training all help your body absorb the work you are doing.

Warm-Up

Every run begins with movement designed to increase blood flow, improve mobility, activate key muscle groups, and prepare the body for running.

Everyday Dynamic Warm-Up

LMLS (Lunge Matrix + Leg Swings)

Lunge Matrix:

- 5 forward lunges each leg
- 5 side lunges each leg
- 5 reverse lunges with rotation each leg

Leg Swings:

- 5 front-to-back each leg
- 5 side-to-side each leg

ORAH (Arm Huggers + Overhead Reaches)

Arm Swings (AH)

- Cross your arms in front of your chest, then open them wide, allowing the shoulder blades to come together. Repeat in a smooth, controlled motion to loosen the shoulders and upper back.

Overhead Reaches (OR)

- Raise one arm overhead and bend it behind your head. Use your opposite hand to gently push the raised arm farther behind your head until you feel a light stretch through the shoulder and triceps. Repeat on both sides while keeping the movement controlled and avoiding any pain.

Workout Warm-Up

Before workouts, athletes should complete 10–15 minutes of easy running, keeping the effort controlled and the heart rate generally below 180 minus age.

Running Drills

After the warm-up, athletes perform short drills such as 2x A-skips, B-skips, high knees, butt kicks, carioca in both directions, Frankensteins, and backwards running. The goal is not conditioning; it is improving movement quality and coordination.

Strides

Strides are one of the most important parts of training. They are fast, relaxed running, not sprinting. Think of them as running around current mile-race effort while staying smooth and controlled.

Focus on:

- Tall posture
- High cadence
- Powerful arm drive (backwards)
- Relaxed shoulders
- Good knee lift
- Running underneath your center of mass

On a track, we typically do $8 \times 100\text{m}$, jogging the curves and striding the straights. On roads or trails, $6\text{--}8 \times 100\text{ m}$ with walk-back recovery works well.

Hill Strides

Hill strides are one of our favorite workouts because they improve running economy, power, mechanics, and strength at the same time.

We typically perform 10–20 repetitions of 20–25 seconds on a moderate hill with jog recovery back down. Zig-zagging downhill helps reduce stress on the quads.

Focus on driving the knees upward, pushing the arms down the slope (backwards), keeping the hips tall, taking short quick steps, and staying relaxed through the shoulders. Mechanics matter more than speed.

Tempo Runs

Tempo runs raise lactate threshold and teach athletes to run faster while remaining primarily aerobic. Typical tempos range from 2–5 miles.

A useful guideline is $\text{current 5K pace} \div 0.93$. For example, a 16:00 5K corresponds to roughly 5:30 pace for tempo running.

Tempo effort should feel comfortably hard, controlled, and sustainable, never all-out.

VO₂ Max Intervals

VO₂ max workouts increase aerobic power and improve the ability to run hard for longer periods.

Advanced runners typically do $5\text{--}8 \times 1000\text{ m}$ at current 5K pace with 2:00 400m jog recovery, while less experienced runners usually do $5\text{--}8 \times 800\text{ m}$ at the same relative effort.

The goal is consistency. The first repetition should not be the fastest.

Long Runs

Long runs build endurance more effectively than any other run in the week. Advanced athletes generally work toward 75-minute medium-long runs and 90-minute long runs.

The focus is relaxed running, good rhythm, and time on your feet rather than pace.

Shakeouts

A shakeout is not another workout. Its purpose is to increase blood flow, promote recovery, and accumulate aerobic volume with very little stress.

Advanced athletes who double may begin doing shakeouts after a long run: 15 minutes to start, adding 3 minutes per week until they reach 30 minutes. Shakeouts should feel easier than an easy run.

Strength Training

Running builds the engine; strength builds the chassis. We encourage core, glute, hip, and stability work at least 2–3 times per week, and ideally some form of activation or mobility most days.

Strong runners tend to stay healthy longer.

How We Progress

Training progresses in four ways: more time, better quality, more consistency, and smarter recovery.

Runs gradually become longer, workouts become more demanding, athletes miss fewer days, and experienced runners learn how to recover more effectively.

Advanced Training Guidelines

Varsity athletes generally work toward at least 60 minutes of running every day, a 75-minute medium-long run, a 90-minute long run, and no more than two quality workouts per week.

They may add shakeout doubles after long-run days, run 30–60 minutes easy after races, and complete a short shakeout the morning before evening races.

The goal is to feel fresh 2–3 days before races, keep easy days truly easy, maintain mileage through most of the season, and only significantly reduce volume for the championship taper.

Final Thoughts

Remember these principles throughout the season:

- Train consistently before you train hard.
- Easy days should feel easy.
- Hard workouts should have a purpose.
- Never race your teammates in practice.
- Good mechanics matter more than fast splits.
- Listen to your body, but do not confuse discomfort with injury.
- Fitness is built over months, not days.
- Recovery is part of training.
- Trust the process.

The Most Important Idea

We train adaptations, not workouts.

Every session exists to create a specific adaptation. After a workout, ask yourself: What was today trying to improve? Why was this pace chosen? Why was this recovery chosen? When athletes understand why they are training, they become smarter, more independent runners.